

Project Report – Phase I

Design of a Graph-Based Academic Library Management System

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Abstract

This project focuses on the design of a Graph-Based Academic Library Management System using PostgreSQL. Traditional library systems typically store entities such as authors, publications, journals, and institutions in separate relational tables. While this structure works well for simple queries, it becomes difficult to analyze complex relationships between academic entities.

To address this limitation, this project adopts a graph-based approach where entities are modeled as nodes and relationships between them are modeled as edges. This design allows the system to represent and analyze connections such as co-author collaborations, citation relationships between publications, journal influence networks, and institutional affiliations.

The system will simulate graph database behavior using PostgreSQL while maintaining relational database capabilities. By storing entities and relationships in a graph-like structure, the system can support network analysis and advanced academic research insights.

Introduction

Academic research data contains many interconnected relationships. Authors collaborate with each other, publications cite earlier work, journals influence research impact, and institutions connect researchers through affiliations.

Traditional relational database systems represent these entities in multiple tables, which makes it difficult to explore relationship-based queries such as:

- Who collaborates with a specific author?
- Which publications cite each other?
- Which authors publish in high-impact journals?

- How research influence spreads through collaborations?

To better represent these relationships, this project designs a graph-based academic library system.

In a graph model:

- Entities are represented as nodes
- Relationships between entities are represented as edges
- This approach makes it easier to analyze networks such as:
- Author collaboration networks
- Publication citation networks
- Journal influence networks
- Institutional affiliation networks

The system will be implemented using PostgreSQL as the database, Python and Flask as the backend, and NetworkX with Matplotlib for graph visualization.

System Concept

The Academic Library Management System models academic entities and their relationships using a graph-oriented data structure.

The main entities in the system include:

- Authors
- Publications
- Journals
- Institutions
- Library Members

Each entity is represented as a node, while interactions between them are represented as relationships (edges).

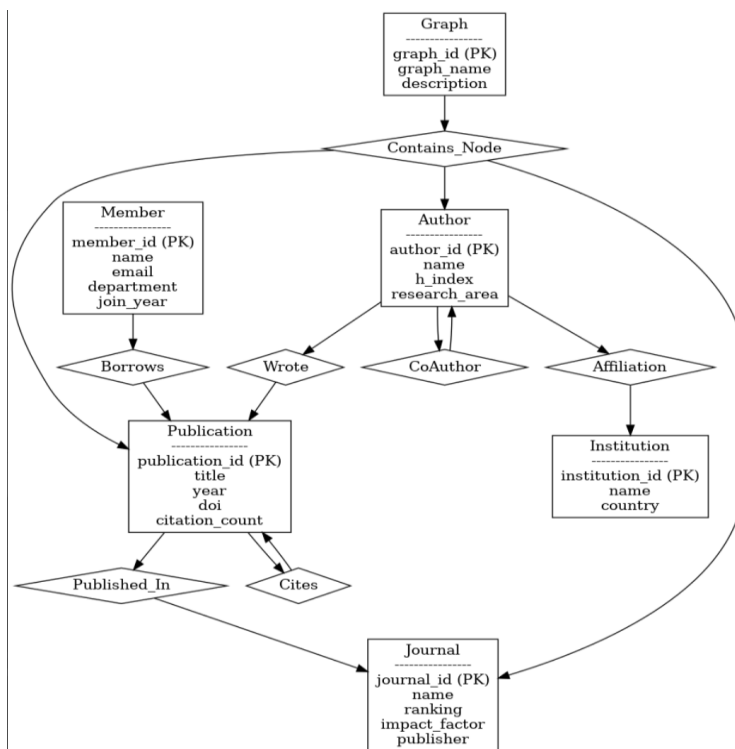
For example:

- An Author writes a Publication
- A Publication is published in a Journal
- A Publication cites another Publication
- An Author collaborates with another Author
- An Author belongs to an Institution

- A Member borrows a Publication

This design allows the system to perform network-based analysis and visualize academic relationships more effectively.

ER Diagram Design



The ER diagram represents the structure of the graph-based academic library system. It defines the major entities, their attributes, and the relationships between them.

The key entities in the system are:

Graph

Represents a specific graph network used for analysis.

Attributes:

- graph_id (Primary Key)
- graph_name
- description

Author

Represents a researcher who writes academic publications.

Attributes:

- author_id (Primary Key)
- name
- h_index
- research_area

Publication

Represents academic research papers, articles, or books.

Attributes:

- publication_id (Primary Key)
- title

- year
- doi
- citation_count

Journal

Represents academic journals where publications are published.

Attributes:

- journal_id (Primary Key)
- name
- ranking
- impact_factor
- publisher

Institution

Represents universities or research organizations affiliated with authors.

Attributes:

- institution_id (Primary Key)
- name
- country

Member

Represents library users who can borrow publications.

Attributes:

- member_id (Primary Key)
- name
- email
- department
- join_year

System Architecture

The system will be implemented using a three-layer architecture.

Database Layer

The database layer uses PostgreSQL to store entities and relationships.

Responsibilities include:

- Storing nodes and edges
- Maintaining relationships between entities

- Executing SQL queries

Backend Layer

The backend will be implemented using Python and Flask.

Responsibilities include:

- Connecting the frontend with the database
- Running graph queries
- Processing results
- Returning graph data for visualization

Visualization Layer

The system will use NetworkX and Matplotlib to visualize graph networks.

Responsibilities include:

- Constructing graph structures
- Displaying relationships visually
- Analyzing collaboration networks and citation graph

Technology Stack

The project will use the following technologies:

- Technology, Purpose
- PostgreSQL, Database storage
- Python, Backend programming
- Flask, Web application framework
- NetworkX, Graph analysis
- Matplotlib, Graph visualization
- HTML, Frontend structure
- Bootstrap, Frontend styling

PostgreSQL is chosen because it supports recursive queries and complex joins, which are useful for graph-like data analysis.

Development Timeline

The project development is divided into four phases.

Phase I – Conceptual Design and ER Modeling

- Define system requirements
- Design ER diagram
- Identify entities and relationships

Phase II – Database Schema Implementation

Create database schema

- Implement tables
- Insert sample academic data
- Expected completion: March 31

Phase III – Graph Query Implementation

- Implement co-author network query
- Implement h-index calculation
- Implement journal influence network
- Build graph visualization
- Expected completion: April 15

Phase IV – Final System and Demonstration

- Complete system integration
- Perform testing
- Prepare final presentation
- Expected completion: May 1

Conclusion

This project introduces a Graph-Based Academic Library Management System that models academic data as interconnected nodes and relationships. By adopting a graph-oriented design, the system allows researchers and users to explore complex academic relationships such as collaborations, citations, and journal influence networks.

The system combines the power of relational databases with graph-based analysis techniques. Using PostgreSQL, Python, Flask, and NetworkX, the system demonstrates how traditional databases can simulate graph database behavior.

The design provides a flexible and scalable foundation that can be extended in the future to support additional analytics and research network analysis.